

EXP-3 BYN

April 2, 2025

```
[ ]: !pip install pgmpy
```

```
[11]: import numpy as np
import pandas as pd
from pgmpy.models import BayesianModel
from pgmpy.estimators import MaximumLikelihoodEstimator
from scipy.stats import chi2_contingency
from pgmpy.inference import VariableElimination

def build_bayesian_network_sample(data, sample_size=50, threshold=0.05):
    if sample_size is not None:
        sampled_data = data.sample(n=min(sample_size, len(data)),
        ↪random_state=42) # Sample the data.
    else:
        sampled_data = data

    variable_order = list(sampled_data.columns)
    model = BayesianModel()

    for variable in variable_order:
        model.add_node(variable)
        parents = []
        for potential_parent in model.nodes():
            if potential_parent != variable:
                contingency_table = pd.crosstab(sampled_data[potential_parent],
        ↪sampled_data[variable])
                chi2, p, dof, expected = chi2_contingency(contingency_table)
                if p < threshold:
                    parents.append(potential_parent)
        model.add_edges_from([(parent, variable) for parent in parents])

    model.fit(sampled_data, estimator=MaximumLikelihoodEstimator)
    return model

# Load the heart disease dataset from the CSV file 'heart.csv'.
data = pd.read_csv('heart.csv')
```

```

# Build the Bayesian network with a sample of 50 rows.
model = build_bayesian_network_sample(data, sample_size=50)

# Print the CPDs.
for cpd in model.get_cpds():
    print("CPD for:", cpd.variable)
    print(cpd)
    print("\n")

# Append the inference code.
HeartDisease_infer = VariableElimination(model)

# Corrected query with sex:1
q = HeartDisease_infer.query(variables=['target'], evidence={'age': 37, 'sex': 1})
print(q)

```

WARNING:pgmpy:BayesianModel has been renamed to BayesianNetwork. Please use BayesianNetwork class, BayesianModel will be removed in future.

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```

CPD for: age
+-----+-----+
| age(34) | 0.02 |
+-----+-----+
| age(40) | 0.06 |
+-----+-----+
| age(41) | 0.02 |
+-----+-----+
| age(43) | 0.02 |
+-----+-----+
| age(44) | 0.06 |
+-----+-----+
| age(45) | 0.04 |
+-----+-----+
| age(46) | 0.06 |
+-----+-----+
| age(48) | 0.02 |
+-----+-----+
| age(50) | 0.02 |
+-----+-----+
| age(51) | 0.04 |
+-----+-----+
| age(52) | 0.08 |
+-----+-----+

```

age(54) 0.06
+-----+-----+
age(55) 0.02
+-----+-----+
age(56) 0.06
+-----+-----+
age(57) 0.1
+-----+-----+
age(58) 0.02
+-----+-----+
age(59) 0.08
+-----+-----+
age(61) 0.04
+-----+-----+
age(62) 0.04
+-----+-----+
age(63) 0.02
+-----+-----+
age(64) 0.04
+-----+-----+
age(66) 0.02
+-----+-----+
age(67) 0.02
+-----+-----+
age(71) 0.04
+-----+-----+

CPD for: sex

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
age age(34) age(40) age(41) age(43) ... age(63) age(64)
age(66) age(67) age(71)
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
sex(0) 1.0 0.0 0.0 0.0 ... 0.0 0.0 1.0
0.0 1.0
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
sex(1) 0.0 1.0 1.0 1.0 ... 1.0 1.0 0.0
1.0 0.0
+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+

CPD for: cp

+-----+-----+
cp(0) 0.52

```

+-----+-----+
| cp(1) | 0.16 |
+-----+-----+
| cp(2) | 0.2  |
+-----+-----+
| cp(3) | 0.12 |
+-----+-----+

```

CPD for: trestbps

```

+-----+-----+
| trestbps(104) | 0.02 |
+-----+-----+
| trestbps(105) | 0.02 |
+-----+-----+
| trestbps(110) | 0.06 |
+-----+-----+
| trestbps(115) | 0.02 |
+-----+-----+
| trestbps(118) | 0.04 |
+-----+-----+
| trestbps(120) | 0.1  |
+-----+-----+
| trestbps(122) | 0.02 |
+-----+-----+
| trestbps(124) | 0.02 |
+-----+-----+
| trestbps(125) | 0.04 |
+-----+-----+
| trestbps(128) | 0.08 |
+-----+-----+
| trestbps(130) | 0.06 |
+-----+-----+
| trestbps(132) | 0.02 |
+-----+-----+
| trestbps(134) | 0.02 |
+-----+-----+
| trestbps(138) | 0.06 |
+-----+-----+
| trestbps(140) | 0.14 |
+-----+-----+
| trestbps(150) | 0.08 |
+-----+-----+
| trestbps(152) | 0.02 |
+-----+-----+
| trestbps(160) | 0.06 |
+-----+-----+
| trestbps(165) | 0.02 |

```

```

+-----+-----+
| trestbps(170) | 0.06 |
+-----+-----+
| trestbps(178) | 0.02 |
+-----+-----+
| trestbps(200) | 0.02 |
+-----+-----+

```

CPD for: chol

```

+-----+-----+
| chol(126) | 0.02 |
+-----+-----+
| chol(166) | 0.02 |
+-----+-----+
| chol(167) | 0.02 |
+-----+-----+
| chol(168) | 0.02 |
+-----+-----+
| chol(188) | 0.02 |
+-----+-----+
| chol(192) | 0.02 |
+-----+-----+
| chol(199) | 0.02 |
+-----+-----+
| chol(204) | 0.04 |
+-----+-----+
| chol(205) | 0.02 |
+-----+-----+
| chol(207) | 0.02 |
+-----+-----+
| chol(208) | 0.04 |
+-----+-----+
| chol(210) | 0.02 |
+-----+-----+
| chol(214) | 0.02 |
+-----+-----+
| chol(221) | 0.02 |
+-----+-----+
| chol(223) | 0.04 |
+-----+-----+
| chol(225) | 0.02 |
+-----+-----+
| chol(226) | 0.02 |
+-----+-----+
| chol(227) | 0.02 |
+-----+-----+
| chol(235) | 0.02 |

```

+-----+-----+		
chol(237)	0.02	
+-----+-----+		
chol(242)	0.02	
+-----+-----+		
chol(243)	0.02	
+-----+-----+		
chol(245)	0.02	
+-----+-----+		
chol(247)	0.02	
+-----+-----+		
chol(254)	0.04	
+-----+-----+		
chol(255)	0.02	
+-----+-----+		
chol(260)	0.02	
+-----+-----+		
chol(261)	0.02	
+-----+-----+		
chol(263)	0.04	
+-----+-----+		
chol(265)	0.02	
+-----+-----+		
chol(267)	0.02	
+-----+-----+		
chol(270)	0.02	
+-----+-----+		
chol(273)	0.04	
+-----+-----+		
chol(276)	0.02	
+-----+-----+		
chol(283)	0.02	
+-----+-----+		
chol(286)	0.02	
+-----+-----+		
chol(288)	0.04	
+-----+-----+		
chol(289)	0.04	
+-----+-----+		
chol(302)	0.02	
+-----+-----+		
chol(311)	0.02	
+-----+-----+		
chol(325)	0.02	
+-----+-----+		
chol(409)	0.02	
+-----+-----+		

CPD for: fbs

```
+-----+-----+
| fbs(0) | 0.76 |
+-----+-----+
| fbs(1) | 0.24 |
+-----+-----+
```

CPD for: restecg

```
+-----+-----+-----+-----+-----+
| cp      | cp(0)          | cp(1) | cp(2) | cp(3)          |
+-----+-----+-----+-----+-----+
| restecg(0) | 0.6923076923076923 | 0.125 | 0.5   | 0.6666666666666666 |
+-----+-----+-----+-----+-----+
| restecg(1) | 0.3076923076923077 | 0.875 | 0.5   | 0.3333333333333333 |
+-----+-----+-----+-----+-----+
```

CPD for: thalach

```
+-----+-----+
| thalach(71) | 0.02 |
+-----+-----+
| thalach(99) | 0.02 |
+-----+-----+
| thalach(103) | 0.02 |
+-----+-----+
| thalach(105) | 0.02 |
+-----+-----+
| thalach(112) | 0.02 |
+-----+-----+
| thalach(113) | 0.02 |
+-----+-----+
| thalach(114) | 0.04 |
+-----+-----+
| thalach(116) | 0.02 |
+-----+-----+
| thalach(120) | 0.02 |
+-----+-----+
| thalach(124) | 0.02 |
+-----+-----+
| thalach(125) | 0.04 |
+-----+-----+
| thalach(130) | 0.02 |
+-----+-----+
| thalach(133) | 0.02 |
+-----+-----+
| thalach(138) | 0.02 |
```

```

+-----+-----+
| thalach(140) | 0.02 |
+-----+-----+
| thalach(143) | 0.02 |
+-----+-----+
| thalach(145) | 0.04 |
+-----+-----+
| thalach(146) | 0.02 |
+-----+-----+
| thalach(147) | 0.02 |
+-----+-----+
| thalach(148) | 0.04 |
+-----+-----+
| thalach(149) | 0.02 |
+-----+-----+
| thalach(150) | 0.02 |
+-----+-----+
| thalach(152) | 0.04 |
+-----+-----+
| thalach(155) | 0.02 |
+-----+-----+
| thalach(156) | 0.02 |
+-----+-----+
| thalach(159) | 0.04 |
+-----+-----+
| thalach(162) | 0.02 |
+-----+-----+
| thalach(164) | 0.02 |
+-----+-----+
| thalach(166) | 0.02 |
+-----+-----+
| thalach(168) | 0.02 |
+-----+-----+
| thalach(169) | 0.02 |
+-----+-----+
| thalach(172) | 0.04 |
+-----+-----+
| thalach(173) | 0.04 |
+-----+-----+
| thalach(174) | 0.02 |
+-----+-----+
| thalach(175) | 0.02 |
+-----+-----+
| thalach(178) | 0.02 |
+-----+-----+
| thalach(180) | 0.02 |
+-----+-----+
| thalach(181) | 0.02 |

```



```

+-----+-----+
| thalach(184) | 0.02 |
+-----+-----+
| thalach(185) | 0.02 |
+-----+-----+
| thalach(186) | 0.02 |
+-----+-----+
| thalach(192) | 0.02 |
+-----+-----+

```

CPD for: exang

```

+-----+-----+-----+-----+-----+
| cp      | cp(0)          | cp(1) | cp(2) | cp(3)          |
+-----+-----+-----+-----+-----+
| exang(0) | 0.3076923076923077 | 0.875 | 1.0   | 0.8333333333333334 |
+-----+-----+-----+-----+-----+
| exang(1) | 0.6923076923076923 | 0.125 | 0.0   | 0.16666666666666666 |
+-----+-----+-----+-----+-----+

```

CPD for: oldpeak

```

+-----+-----+-----+-----+-----+
-----+
| age      | age(34)          | ... | age(71)          | age(71)
|
+-----+-----+-----+-----+-----+
-----+
| trestbps | trestbps(104)    | ... | trestbps(178)    |
trestbps(200) |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.0) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.1) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.2) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.3) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+

```

```

| oldpeak(0.4) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.5) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.6) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.7) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(0.8) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(1.0) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(1.4) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(1.6) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(1.8) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(1.9) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(2.0) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+
| oldpeak(2.4) | 0.043478260869565216 | ... | 0.043478260869565216 |
0.043478260869565216 |
+-----+-----+-----+-----+-----+
-----+

```

oldpeak(2.6)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(2.8)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(3.0)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(3.2)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(3.6)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(4.0)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak(4.2)	0.043478260869565216	...	0.043478260869565216	
0.043478260869565216				
+-----+-----+-----+-----+-----+				
-----+				

CPD for: slope

+-----+-----+-----+-----+-----+				
-----+				
cp	cp(0)	cp(0)	...	cp(3)
cp(3)				
+-----+-----+-----+-----+-----+				
-----+				
exang	exang(0)	exang(0)	...	exang(1)
exang(1)				
+-----+-----+-----+-----+-----+				
-----+				
oldpeak	oldpeak(0.0)	oldpeak(0.1)	...	oldpeak(4.0)
oldpeak(4.2)				
+-----+-----+-----+-----+-----+				
-----+				
slope(0)	0.0	0.3333333333333333	...	0.3333333333333333
0.3333333333333333				
+-----+-----+-----+-----+-----+				

```

-----+
| slope(1) | 0.0          | 0.3333333333333333 | ... | 0.3333333333333333 |
0.3333333333333333 |
+-----+-----+-----+-----+-----+
-----+
| slope(2) | 1.0          | 0.3333333333333333 | ... | 0.3333333333333333 |
0.3333333333333333 |
+-----+-----+-----+-----+-----+
-----+

```

CPD for: ca

```

+-----+-----+-----+-----+-----+
-----+
| trestbps | trestbps(104) | trestbps(105) | ... | trestbps(170)      |
trestbps(178) | trestbps(200) |
+-----+-----+-----+-----+-----+
-----+-----+
| ca      (0) | 1.0          | 1.0          | ... | 0.6666666666666666 | 1.0
| 0.0          |
+-----+-----+-----+-----+-----+
-----+-----+
| ca      (1) | 0.0          | 0.0          | ... | 0.0          | 0.0
| 0.0          |
+-----+-----+-----+-----+-----+
-----+-----+
| ca      (2) | 0.0          | 0.0          | ... | 0.3333333333333333 | 0.0
| 1.0          |
+-----+-----+-----+-----+-----+
-----+-----+
| ca      (3) | 0.0          | 0.0          | ... | 0.0          | 0.0
| 0.0          |
+-----+-----+-----+-----+-----+
-----+-----+
| ca      (4) | 0.0          | 0.0          | ... | 0.0          | 0.0
| 0.0          |
+-----+-----+-----+-----+-----+
-----+

```

CPD for: thal

```

+-----+-----+-----+-----+-----+
| cp      | cp(0)          | cp(1) | cp(2) | cp(3)          |
+-----+-----+-----+-----+-----+
| thal(0) | 0.038461538461538464 | 0.0   | 0.0   | 0.0          |
+-----+-----+-----+-----+-----+
| thal(1) | 0.11538461538461539 | 0.0   | 0.0   | 0.0          |
+-----+-----+-----+-----+-----+

```

thal(2)	0.3076923076923077	0.875	0.9	0.3333333333333333
-----+	-----+	-----+	-----+	-----+
thal(3)	0.5384615384615384	0.125	0.1	0.6666666666666666
-----+	-----+	-----+	-----+	-----+

CPD for: target

-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
cp	cp(0)	cp(0)	cp(0)	cp(0)	...	cp(3)	cp(3)
cp(3)	cp(3)	cp(3)					
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
exang	exang(0)	exang(0)	exang(0)	exang(0)	...	exang(1)	
exang(1)	exang(1)	exang(1)	exang(1)				
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
slope	slope(0)	slope(0)	slope(0)	slope(0)	...	slope(1)	
slope(2)	slope(2)	slope(2)	slope(2)				
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
thal	thal(0)	thal(1)	thal(2)	thal(3)	...	thal(3)	
thal(0)	thal(1)	thal(2)	thal(3)				
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
target(0)	0.5	0.5	0.5	0.5	...	0.5	0.5
0.5	0.5	0.0					
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
target(1)	0.5	0.5	0.5	0.5	...	0.5	0.5
0.5	0.5	1.0					
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+

CPD for: age_group

-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
age	age(34)	age(40)	age(41)	...	age(63)	age(64)	
age(66)	age(67)	age(71)					
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
age_group(mid_old)	0.0	0.0	0.0	...	0.0	0.0	
0.0	0.0	0.0					
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
-----+	-----+	-----+	-----+	-----+	-----+	-----+	-----+
age_group(mid_young)	0.0	0.0	1.0	...	0.0	0.0	
0.0	0.0	0.0					

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
| age_group(old)      | 0.0   | 0.0   | 0.0   | ... | 1.0   | 1.0   |
1.0   | 1.0   | 1.0   |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
| age_group(young)    | 1.0   | 1.0   | 0.0   | ... | 0.0   | 0.0   |
0.0   | 0.0   | 0.0   |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+

```

```

+-----+-----+
| target  | phi(target) |
+=====+=====+
| target(0) | 0.4783 |
+-----+-----+
| target(1) | 0.5217 |
+-----+-----+

```

[]: